

Aaron Detre

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Education

Northeastern University, Khoury College of Computer Sciences | Boston, MA
B.S. Computer Science, Minor in Psychology, summa cum laude | April 2026 | **GPA: 3.91**

Skills

Languages: Java, TypeScript, JavaScript, SQL, HTML, CSS
Frameworks: React, Next.js, Angular, Spring Boot, Node.js, Express.js
Tools & Platforms: Git, GitHub Actions, Docker, AWS CodePipeline, MongoDB
Operating Systems: Linux (Ubuntu), Windows
Engineering Practices: Agile, CI/CD, Pair Programming, Unit Testing

Experience

Full-Stack Software Developer Co-op (Angular, Typescript, Java Spring Boot)

Technology Enhanced Learning in Science, UC Berkeley Jan. 2025 - June 2025

- Developed production features in an existing Angular codebase for WISE, a globally used, open-source virtual learning platform supporting K–12 science education and ongoing research.
- Collaborated directly with education researchers and classroom teachers to translate research goals and instructional needs into software solutions used in real classrooms.
- Designed and implemented guest-accessible units with registration-free access, allowing WISE to function as a survey tool for studies requiring more flexible, activity-based data collection.
- Built teacher-facing summary visualizations that aggregate class-wide student responses, including a summary display for an AI chat activity highlighting the most and least common ideas detected, making important but underrepresented ideas more visible to instructors and researchers.
- Improved developer workflows by automating internationalization annotation generation using GitHub Actions, reducing manual effort and preventing recurring errors in pull requests.
- Wrote unit tests, participated in peer code reviews, and iterated based on feedback from engineers.

Projects

Recipe Social Network (React, Next.js, Node.js, MongoDB)

Nov. 2025 - Dec. 2025

- Designed and built a full-stack recipe-sharing platform with authentication, allowing users to create, review, and remix recipes while following others for a personalized feed.
- Implemented scalable ingredient handling in response to user feedback, supporting numeric and non-numeric quantities and dynamic scaling via user-defined multipliers.
- Developed non-relational MongoDB schemas and implemented Express.js REST endpoints to support search functionality across both internal data and an external recipe API.

Programming Language Implementation (Java, JUnit)

Sept. 2025 - Dec. 2025

- Designed and implemented a 12,000-line object-oriented Java system for a module-based programming language, including an abstract syntax tree, parser, semantic validity checker, and CESK machine.
- Implemented support for programs combining typed and untyped modules with cross-module imports, enforcing soundness via runtime proxies and ensuring safe interactions at module boundaries.
- Applied the Strategy Pattern to improve maintainability by modularizing 25 CESK transition cases.
- Developed a comprehensive testing suite with 100+ integration tests written in the target language alongside JUnit tests, with automated execution via a Bash-based test runner.

Interests

Urban and Nature Photography | Independent Films | Bread Baking